

KUSHAL KUMAR B

B.Tech — Artificial Intelligence & Machine Learning | PES University, Bangalore
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OBJECTIVE

Motivated and organized Computer Science Engineering student specializing in Artificial Intelligence and Machine Learning, with hands-on experience in software development, machine learning projects, and an internship at Hindustan Aeronautics Limited (HAL). Seeking a challenging entry-level role to apply technical skills, problem-solving abilities, and innovative thinking.

EDUCATION

PES University, RR Campus, Bangalore | Expected Graduation: 2027

B.Tech in Artificial Intelligence & Machine Learning (AIML) — Pursuing

Government Polytechnic, Channasandra, Bangalore | 2024

Diploma in Computer Science — Distinction

Amara Jyothi English School | 2021

10th Grade — 76.48%

TECHNICAL SKILLS

Programming: Python, C, Basics of Java

AI/ML Libraries: LangChain, HuggingFace, FAISS, Scikit-learn, OpenCV, MediaPipe, Kivy, Ollama, TensorFlow/Keras

Computer Vision: MediaPipe, OpenCV, Gesture Recognition, Real-time video processing

GUI Development: PyQt5, Kivy

Database: SQL, DBMS fundamentals, FAISS Vector Store

Web Development: HTML, JavaScript, MERN Stack

Software Tools: Microsoft Word, Excel, PowerPoint

Operating Systems: Windows, Linux

Other Skills: Software troubleshooting, technical problem-solving

INTERNSHIP

Hindustan Aeronautics Limited (HAL) — Intern

- Gained practical exposure to industrial software systems and operational workflows.
- Assisted in analyzing and troubleshooting software-related tasks.
- Enhanced problem-solving and collaborative skills in a professional engineering environment.

PROJECTS

AI-Based Hand Gesture Control System with Air Writing | Python, OpenCV, MediaPipe, PyQt5, TensorFlow/Keras, PyAutoGUI, Pytesseract

- Built a real-time hand gesture recognition system using MediaPipe and OpenCV to control mouse movements, clicks, zoom, and screen navigation via webcam — no physical input device required.
- Implemented an Air Writing mode enabling users to write in the air using their index finger; strokes are drawn on a live canvas overlay, then processed and recognized using Pytesseract OCR with optional Keras model support.
- Designed gesture actions including tap-to-click, pinch-to-zoom, thumb swipe navigation (left/right/scroll), and pinch-to-submit / pinch-to-erase for air writing, all with configurable cooldowns and thresholds.
- Developed a full PyQt5 GUI with live webcam feed, real-time FPS display, handedness detection, and adjustable sliders for tap threshold, zoom sensitivity, and movement smoothing.

LLM-RAG — Offline Document Chatbot | *Python, LangChain, FAISS, HuggingFace, Ollama (LLaMA 3)*

- Built a fully offline Retrieval-Augmented Generation (RAG) chatbot enabling natural language querying of PDF documents without internet connectivity.
- Implemented PDF ingestion and chunking pipeline using PyPDF2 and LangChain's RecursiveCharacterTextSplitter (500-token chunks, 100-token overlap).
- Integrated HuggingFace's all-MiniLM-L6-v2 embedding model with FAISS vector store for fast semantic similarity search and context retrieval.
- Deployed Ollama-hosted LLaMA 3 as the local LLM, constructing context-aware prompts from retrieved chunks for accurate, grounded answers.

Face Recognition-Based Student Attendance System | *Python, Kivy, OpenCV, Scikit-learn*

- Developed a Kivy-based desktop application to automate student attendance using real-time facial recognition.
- Implemented and benchmarked ML classifiers (Random Forest, SVM, CNN) for high recognition accuracy.
- Integrated live recognition with a database backend to automatically log student names, dates, and timestamps.

Semantic Search Application | *Python, NLP*

- Created a semantic search engine that understands contextual meaning in natural language queries.
- Utilized NLP techniques to deliver significantly better accuracy than traditional keyword-based systems.
- Designed for improved document retrieval in educational and research datasets.

LANGUAGES

English, Kannada, Hindi, Telugu

HOBBIES & INTERESTS

Competitive coding on platforms like HackerRank; Building AI & ML personal projects such as motion detection, voice assistants, face recognition apps, hand gesture control systems, and offline LLM-powered document tools.